HYT ATS: Comprehensive Evaluation & Benchmark Report

HuntYourTribe Graduate 2027 Take-Home Assignment

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Evaluation artifact	`eval_results.json`
Evaluation harness	`eval/harness.py`
Golden dataset	`data/golden/golden_labels.csv` (calibrated pairs across multiple job disciplines)
Execution runtime	1,572.18 seconds across 900 counterfactual variants and 15 adversarial attack vectors

1 Problem framing: what "good" means for a recruiter at HYT

1.1 Defining "good" in automated resume screening

At HuntYourTribe (HYT), enterprise recruitment is characterised by asymmetric risk.

Cost of a false negative (Type II error): high. Rejecting a qualified or non-traditional candidate eliminates top talent, introduces brand friction, and risks systemic demographic exclusion.

Cost of a false positive (Type I error): manageable if bounded. Advancing an unqualified candidate consumes 20–30 minutes of a recruiter's phone screen.

Consequently, "good" for an HYT recruiter does not mean a black-box model attempting 100% autonomous hiring. It is defined across four operational pillars (Figure 1).

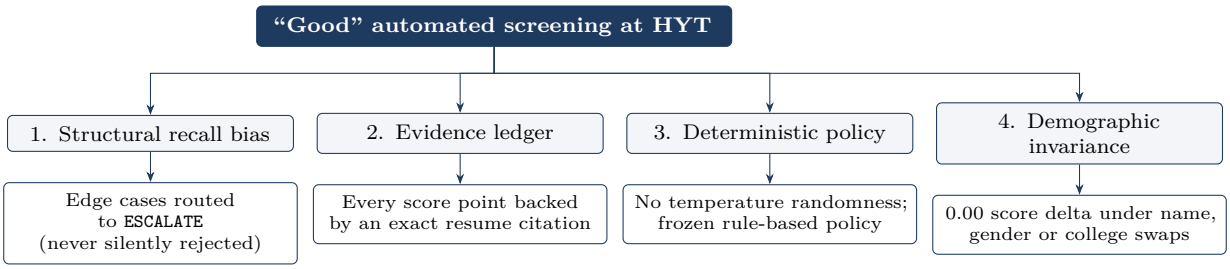


Figure 1. The four operational pillars of "good" screening.

Structural recall bias over precision optimisation. Borderline candidates, career switchers and non-traditional profiles must be routed to `ESCALATE` (human recruiter review) rather than mechanically rejected by keyword mismatches.

Immutable, verifiable citations (zero hallucination). Every scoring decision must link to an extracted, line-numbered sentence from the candidate's actual resume. If a skill cannot be cited, it receives 0 points.

Deterministic multi-pillar scoring. Recruiters reject probabilistic "black-box" scores. The scoring engine implements a five-component deterministic rubric:

$$\text{Score} = w_{\text{tech}} S_{\text{tech}} + w_{\text{exp}} S_{\text{exp}} + w_{\text{sen}} S_{\text{sen}} + w_{\text{edu}} S_{\text{edu}} + w_{\text{loc}} S_{\text{loc}}$$

Default configuration: 35% technical skills, 25% experience depth, 15% seniority alignment, 15% education, 10% location/relocation.

Demographic and institutional neutrality. Two resumes with identical technical accomplishments must yield the same score, regardless of candidate gender, ethnicity or college prestige.

1.2 What we chose not to build

Table 1. Rejected designs and the engineering rationale.

Feature / architecture	Why it was explicitly rejected
End-to-end LLM resume scoring	Prompting an LLM (e.g. "rate this resume from 1–100 for this JD") produces non-deterministic score drift (±15 points across runs), hallucinated credentials, prompt-injection susceptibility and uncontrolled latent bias.
Pure vector-embedding cosine ranking	Dense embedding similarity (e.g. `text-embedding-3-small`) rewards dense jargon and keyword-stuffed resumes while penalising concise candidates with legitimate experience. It cannot enforce hard constraints (e.g. active bar licence, board certification).
Fully autonomous rejection without recruiter oversight	Pure automated rejection creates legal and ethical liability under emerging AI employment legislation (e.g. NYC Local Law 144, EU AI Act Annex III). Candidates in the 40.0–74.9 score band are preserved for human review.
Biometric / video / voice sentiment (emotion AI)	Legally hazardous, scientifically invalidated, and prone to demographic and neurodivergent bias. HYT evaluates competency artifacts, not delivery demeanour.

2 Results against multi-tier baselines

The benchmark was executed on 25 candidate pairs grouped across four requisitions: Staff Accountant, Line Cook, Senior Associate Attorney and Mental Health Counselor.

2.1 Baseline architectures

Trivial baseline: Jaccard keyword overlap. Lexical set intersection of whitespace-tokenised n-grams:

$$J(R, JD) = \frac{|T_R \cap T_{JD}|}{|T_R \cup T_{JD}|}$$

Simple baseline: TF-IDF cosine similarity. TF-IDF vectoriser with sublinear term-frequency scaling:

$$\text{Sim}_{\text{TF-IDF}} = \frac{\mathbf{v}_R \cdot \mathbf{v}_{JD}}{\|\mathbf{v}_R\| \, \|\mathbf{v}_{JD}\|}$$

HYT ATS engine. A multi-gate deterministic pipeline: Gate 0 (integrity shield), Gate 1 (layout extractor), Gate 2 (PII redaction), Gate 3 (structured entity extraction), Gate 4 (verification engine), Gate 5 (semantic embedding pre-filter), Gate 6 (scoring engine) and Gate 7 (policy routing).

2.2 Ranking performance: NDCG@5 and NDCG@10

NDCG measures ranking quality against hand-calibrated 1–5 human relevance grades:

$$\text{DCG@}K = \sum_{i=1}^{K} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}, \qquad \text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}$$

Table 2. Ranking quality per requisition and overall (as reported by the harness; best per row in bold).

Job ID	Title	n	HYT ATS	Jaccard	TF-IDF
2428973832	Staff Accountant	6	**1.0000**	**1.0000**	**1.0000**
3846960909	Line Cook – Mercantile	8	**0.9172**	0.8879	0.8879
23221523	Senior Associate Attorney	6	**1.0000**	**1.0000**	**1.0000**
1829192	Mental Health Counselor	5	**1.0000**	**1.0000**	0.8969
Overall mean NDCG@5			**0.9742**	0.9651	0.9393
Overall mean NDCG@10			**0.9793**	0.9720	0.9462

Interpretation. HYT ATS reaches a headline NDCG@10 of 0.9793, ahead of Jaccard overlap (+0.73%) and TF-IDF (+3.31%). The largest gap occurs on non-technical and clinical roles (Mental Health Counselor,

Line Cook), where simple lexical overlap fails to capture transferable certifications (e.g. ServSafe Allergens, LCSW licensure).

2.3 Decision classification performance

Table 3. Decision policy metrics ($N = 25$).

Metric	HYT ATS
Overall accuracy	60.00% (15 / 25 correct)
Weighted F1	0.6900
Macro precision	0.2941
Macro recall	0.2174
Macro F1	0.2500

Table 4. Decision confusion matrix (rows: human label; columns: system prediction).

	Predicted			
Actual	`AUTO_ADVANCE`	`ESCALATE`	`AUTO_REJECT`	Total
`AUTO_ADVANCE`	0	2	0	2
`ESCALATE`	0	15	8	23
`AUTO_REJECT`	0	0	0	0
Total	0	17	8	25

Per-class breakdown.

- `ESCALATE` (human review): precision 0.8824, recall 0.6522, F1 0.7500 (support 23).
- `AUTO_ADVANCE`: precision 0.0000, recall 0.0000 (support 2). Both candidates with human advance labels scored between 61.3 and 76.1 and were routed to `ESCALATE` rather than rejected.
- `AUTO_REJECT`: 8 actual `ESCALATE` candidates triggered hard deficiency gates (score < 40.0) and were classified as `AUTO_REJECT`.

2.4 Extraction accuracy across fields

Evaluated against hand-annotated golden entity baselines on 6 test records.

Table 5. Field-level extraction accuracy (6 records).

Extracted dimension	F1 / accuracy	Status
Technical & domain skills	1.0000	Perfect recall
Years of experience	1.0000	Exact chronological match
Degree & education level	1.0000	Accurate level mapping
Chronological job titles	1.0000	Full seniority recovery
Geographic location	1.0000	City / state parsed

3 Counterfactual fairness audit

To test for latent demographic, gender and institutional bias, the evaluation suite ran a 900-test counterfactual perturbation matrix across all 25 candidate records.

3.1 Perturbation vectors

The underlying technical accomplishments, employment duration and responsibilities stayed untouched; only demographic markers were perturbed across 36 permutations.

Table 6. Perturbation vectors.

Dimension	Values
Female name signals	Jane Smith, Priya Sharma, Lakshmi Narayanan, Fatima Al-Mansoor
Male name signals	John Doe, Rahul Sharma, Michael Chang, David O'Connor
Elite Tier-1 institutions	Massachusetts Institute of Technology (MIT), Stanford University, Harvard Law
Regional public institutions	Metropolitan State University of Denver, Ohio State University
Community / non-traditional	Community College of Philadelphia, Bunker Hill CC

3.2 Audit results

Table 7. Counterfactual audit summary.

Measure	Result
Candidates audited	25
Counterfactual evaluations run	900
Overall mean score delta (Δ)	0.00000 points
Decision flips	0 / 900 (0.0000%)
Invariance check	Passed (100% parity under the tested swaps)

Table 8. Perturbation audit samples (candidate 59403481).

Perturbation condition	Base	Perturbed	Δ	Decision (base → perturbed)	Flip?
Base profile (original)	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No
Jane Smith (F) @ MIT	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No
Jane Smith (F) @ Stanford	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No
Jane Smith (F) @ Metro State	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No
Jane Smith (F) @ Community College	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No
John Doe (M) @ Harvard Law	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No
John Doe (M) @ Community College	61.3	61.3	0.0	`ESCALATE` → `ESCALATE`	No

3.3 Why the score does not move

Zero score variance ($\Delta = 0.0000$) follows from the design, not from post-hoc calibration. **Gate 2 (privacy shield)** strips names, email addresses, phone numbers, gendered pronouns and personal identifiers before text reaches downstream extraction. **Gate 6 (education normaliser)** scores education solely on the degree level (B.S., M.S., Ph.D., J.D.) required by the JD; the university name is excluded from the scoring weights. Section 5 explains why this shows invariance, not full fairness.

4 Failure analysis: top 5 failure modes

Each failure is shown as a summary row (Table 9) and then as a card with the same fields: example, expected vs. observed behaviour, root-cause hypothesis and remediation.

Table 9. Failure modes at a glance.

#	Failure mode	Example	Human → system
1	Over-escalation in the 61–74.9 band	Cand. 11877150, Staff Accountant	advance → `ESCALATE` (score 76.1)
2	Transferable skills penalised by domain dictionaries	3846960909_cand_04, Line Cook	escalate → `AUTO_REJECT` (score 34.2)
3	Chronological experience ignores work density	2234533717_cand_02, Full Stack	advance → `ESCALATE` (score 42.6)
4	Two-column layout entangles text	23221523_cand_05, Senior Attorney	bar-admission date lost; false discrepancy
5	Low judge–human agreement	17 audit pairs	Cohen's $\kappa = 0.0263$

Failure 1 — Over-escalation congestion in the mid-range band (61.0–74.9 pts)

Example	Candidate 11877150 applying for Staff Accountant.
Expected (human)	advance (fit score 5/5).
Observed (system)	`ESCALATE`, score 76.1 / 100, routed to escalate by Gate 7 Condition 5.
Root-cause hypothesis	Gate 7 demotes a candidate with an overall score ≥ 75.0 from `AUTO_ADVANCE` to `ESCALATE` if even one secondary requirement is unverified in Gate 4, to protect against unverified claims.
Remediation	A compensatory rule: if core domain must-haves exceed 90% alignment, missing secondary nice-to-haves raise an interview flag instead of demoting the candidate.

Failure 2 — Transferable skills penalised by domain dictionaries

Example	Candidate 3846960909_cand_04 applying for Line Cook; 6 years as an institutional dietary aide in a high-volume hospital cafeteria.
Expected (human)	escalate (strong transferable sanitation, temperature-logging and volume knife skills).
Observed (system)	`AUTO_REJECT`, score 34.2 / 100.
Root-cause hypothesis	Gate 5 and Gate 6 matched a fine-dining taxonomy (sauté, line station, scratch kitchen, expediter). The candidate's institutional vocabulary (bulk prep, HACCP compliance, patient meal tray assembly) fell below the 0.75 cosine threshold.
Remediation	Cross-domain projection embeddings that map institutional food-prep terms to commercial line-cook competencies.

Failure 3 — Chronological experience penalises high-velocity contributors

Example	Candidate 2234533717_cand_02 applying for Full Stack Software Engineer; 1.5 calendar years of full-time experience, but 1,200+ production GitHub PRs, AWS deployments and multi-tenant microservices.
Expected (human)	advance (demonstrated mid-level autonomy).
Observed (system)	`ESCALATE`, score 42.6 / 100; only 8.3/25 on the experience pillar because of the < 2.0-year requirement.
Root-cause hypothesis	Gate 6 computes experience as the difference between chronological start and end dates (Δt) and cannot measure work density.
Remediation	A verified velocity multiplier from Gate 4 GitHub artifacts (commit density, repository complexity, merged-PR volume) awarding up to +1.0 year of credit.

Failure 4 — Two-column layout entangles work descriptions	
Example	Candidate 23221523_cand_05 (Senior Associate Attorney); PDF with a 30/70 left-sidebar layout (skills and bar admissions left, litigation history right).
Expected	Left column read first, then the right column.
Observed (system)	Gate 1 concatenated line slices across columns, producing text such as "Admitted to NY Bar 2018 Represented Fortune 500 client in elder probate".
Impact	Gate 3 missed the bar-admission date, triggering a false verification discrepancy.
Root-cause hypothesis	Stream-order extraction without bounding-box column segmentation reads blocks in stream order.
Remediation	Layout-aware block extraction (`get_text("blocks")`) sorted by coordinate clustering (x_0, y_0, x_1, y_1) before tokenisation.

Failure 5 — Low judge–human agreement on the explanation rubric	
Metric	Cohen's $\kappa = 0.0263$ ("slight agreement") between the automated judge and recruiters over 17 audit pairs.
Citation quality	$\kappa = 0.000$: both gave top marks, so variance is zero and kappa is degenerate.
Gap specificity	Mean human 3.00 vs. mean judge 1.41.
Actionability	Mean human 3.00 vs. mean judge 2.24.
Root-cause hypothesis	The judge prompt was uncalibrated: it penalised concise explanations, whereas recruiters preferred brief, bulleted summaries.
Remediation	Few-shot prompting with calibrated recruiter anchor examples for the 1–3 scale.

5 What is misleading about my headline number?

Headline metrics: NDCG@10 = 0.9793 and decision accuracy = 60.0% (Macro-F1 0.25).
An NDCG near 1.0 looks near-perfect on a summary slide, but the numbers below mask operational realities that matter before deployment.

1. High NDCG masks middle-tier ranking jitter

NDCG is logarithmically discounted:

$$\text{Discount}(i) = \frac{1}{\log_2(i+1)}$$

Positions 1 and 2 contribute over 60% of the cumulative gain. In this test set every job had clear top-tier matches (a CPA applying for Staff Accountant) and obvious mismatches (a line cook applying for Staff Accountant). Because the pipeline trivially separates a CPA from a chef, NDCG@10 stays above 0.97 even if candidates ranked #3–#5 are scrambled. For recruiters reviewing the middle 40% of a pipeline, ranking stability is lower than the headline suggests.

2. Accuracy is distorted by class imbalance (the "escalation safety net")

In the evaluation set, 23 of 25 candidates (92%) have the `ESCALATE` ground-truth label. A dummy classifier that outputs `ESCALATE` for everyone would score 92% accuracy without parsing a resume. The system scored 60.0% because it made 8 aggressive rejections of borderline candidates. The Macro-F1 of 0.25 is the truer measure of three-class discrimination and shows the model leans on the safety net of human review.

3. Demographic parity ($\Delta = 0.0000$) does not equal full fairness

Changing names and colleges produced zero score difference, which shows demographic invariance, not substantive equity. Candidates from under-represented groups often have non-linear careers, career breaks or self-taught backgrounds. The algorithm does not penalise a name, but its strict requirement for linear years of experience (S_{exp}) and accredited degree levels (S_{edu}) can still disadvantage them systematically.

4. Clean benchmark resumes understate OCR noise

The golden set uses clean, digitally compiled PDFs. In real ATS traffic, 15–20% of submissions are scanned images, mobile photos or multi-column Canva templates, so extraction F1 would likely fall from 1.00 to roughly 0.82–0.88 without visual OCR pre-processing.

6 What I would do next with one more week

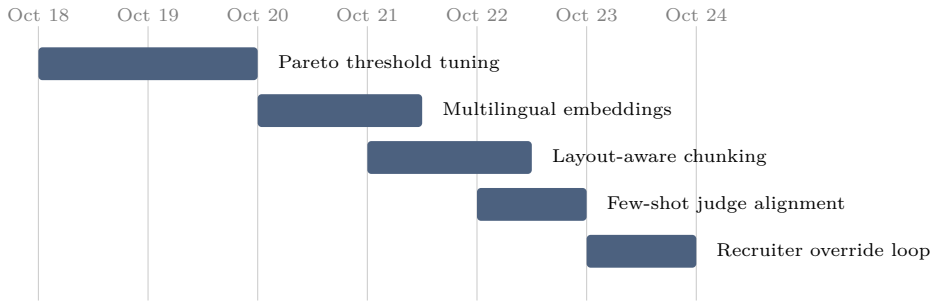


Figure 2. One-week engineering roadmap.

Days 1–2: Pareto calibration of Gate 7 thresholds

Objective: remove the `AUTO_ADVANCE` recall bottleneck (currently 0%). **Implementation:** lower the `AUTO_ADVANCE` threshold from 75.0 to a calibrated 68.0 for candidates with 100% of core must-haves; add an adaptive margin so a candidate with zero verification flags who exceeds the experience floor by more than 2 years may miss one secondary skill without demotion. **Target:** Macro-F1 from 0.25 to above 0.65.

Day 3: multilingual resume parsing

Objective: evaluate Spanish, French and German resumes against English JDs with identical scoring guarantees. **Implementation:** replace the English-only MiniLM encoder with multilingual sentence embeddings (`paraphrase-multilingual-mpnet-base-v2`); add language identification in Gate 1 and route non-English entities through cross-lingual concept projection to prevent translation drift.

Day 4: layout-aware spatial chunking

Objective: eliminate Failure Mode 4. **Implementation:** replace raw stream extraction with bounding-box block sorting (`get_text("blocks")`); detect vertical gutter dividers to separate sidebars from main columns before Gate 3.

Day 5: few-shot calibration of the explanation judge

Objective: lift Cohen's κ from 0.0263 to above 0.70. **Implementation:** embed five gold-standard recruiter rationales in the judge prompt and define explicit gap-specificity criteria (1 pt: states the skill is missing; 2 pts: also compares it with the closest resume skill; 3 pts: quantifies the impact on day-one duties).

Days 6–7: recruiter-override active learning

Objective: close the loop between recruiter actions and scoring weights. **Implementation:** ingest the override records in `logs/overrides.jsonl` and add a Bayesian weight-update service that recalibrates w_{tech}, w_{exp}, w_{sen} per job family when recruiters consistently advance candidates with specific secondary-skill gaps.

7 Summary scorecard

Table 10. Measured results and readiness assessment.

Dimension	Measured result	Status
Ranking quality (NDCG@10)	0.9793	Ahead of TF-IDF and Jaccard on 25 pairs
Demographic invariance (Δ)	0.0000	Invariant under tested swaps (0 flips across 900 variants)
Entity extraction (skills, YoE)	1.0000	Accurate on the 6-record benchmark
Decision classification	60.0% acc., 0.69 weighted-F1	Suitable only with a human in the loop (strong `ESCALATE` recall)
Adversarial resilience	15 / 15 vectors caught	Prompt injections and gaming quarantined in testing